Large-Stepsize GD on Multinomial Logistic Regression: Part I—On Linearly-Separable Datasets

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A New Paradigm for Deep Learning

- Gradient-based optimization frameworks show promises for machine learning.
- Strong theoretical guarantees rely on convexity, smoothness of the objective function, choice of learning rate, etc. - via the descent lemma.

$$\mathcal{L}(\mathbf{w}^{(t+1)}) \leq \mathcal{L}(\mathbf{w}^{(t)}) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla \mathcal{L}(\mathbf{w}^{(t)})\|^2 \overset{\eta<2/L}{<} \mathcal{L}(\mathbf{w}^{(t)}).$$

- Large stepsizes are not suitable for general functions. However, modern machine learning often display oscillating optimization dynamics, causing **unstable convergence**.

Three Pillars	Classical Learning	Modern Answers
Approximation	Non-parametric classes	Depth, Attention, Self-supervision
Optimization	Stable convergence, convexity	Over-parameterization, **Large stepsizes**
Generalization	Model class controls complexity, Early stopping, i.i.d. distribution, Uniform convergence	**Algorithms control complexity**, Benign overfitting, Zero-shot learning

Table: From classical machine learning to modern deep learning (Bartlett, 2025).

The Edge-of-Stability (EoS) Phenomenon

Gradient Descent

$$\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} - \eta \nabla \mathcal{L}(\mathbf{w}^{(k)}), k = 0, 1, 2, \ldots, T, \text{and } \eta > 0$$

For general L-smooth functions $\mathcal{L} \in \mathcal{C}^1$:

- If $\eta \leq 1/L$, then the loss will eventually converge to a local minimum.

Cohen et al. (2021) points out that the classic doctrine of stepsize selection in optimization theory may be unjustified...

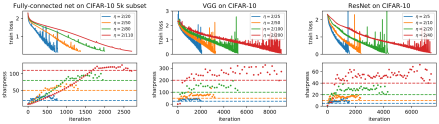


Figure 1: **Gradient descent typically occurs at the Edge of Stability.** On three separate architectures, we run gradient descent at a range of step sizes η, and plot both the train loss (top row) and the sharpness (bottom row). For each step size η, observe that the sharpness rises to $2/\eta$ (marked by the horizontal dashed line of the appropriate color) and then hovers right at, or just above, this value.

Figure: The Edge-of-Stability phenomenon, where the risk oscillates, but still non-monotonically decreases in the long run (Cohen et al., 2021).

A Negative (Yet Motivating) Example

Example

Consider the function

$$f(\theta_1, \theta_2) = \theta_1^2 + 10\theta_2^2 = \frac{1}{2}\begin{bmatrix}\theta_1 \\ \theta_2\end{bmatrix}^\top \begin{bmatrix}2 & 0 \\ 0 & 20\end{bmatrix}\begin{bmatrix}\theta_1 \\ \theta_2\end{bmatrix} =: \frac{1}{2}\boldsymbol{\theta}^\top \mathbf{A}\boldsymbol{\theta}.$$

The smoothness constant is given by $L = 20$, and the minima of f is given by $(\theta_1^\star, \theta_2^\star) = (0, 0)$. Running gradient descent with stepsize η yields $\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^\star = (\mathbf{I} - \eta\mathbf{A})^t \boldsymbol{\theta}^{(0)}$. If any eigenvalue of $\mathbf{A}$ exceeds $2/\eta$, then the sequence $\{\boldsymbol{\theta}^{(t)}\}$ will oscillate around $\boldsymbol{\theta}^\star$ with increasing magnitude and thus diverge.

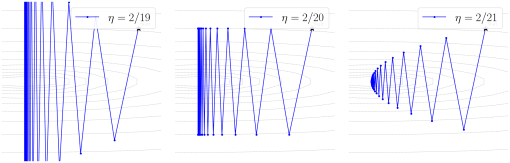


Figure: Optimization trajectory for $f(\theta_1, \theta_2) = \theta_1^2 + 10\theta_2^2$ with $\eta = 2/19$, $2/20$, $2/21$, demonstrating GD behavior near the stability boundary.

Several Lines of Previous Works

- Ahn et al. (2022) states that (almost surely,) **GD cannot converge to a stationary point $\mathrm{w}_\star$ with sharpness greater than $2/\eta$**. The oscillation helps GD to find **flatter minima**.
- Damian et al. (2023) formalized that **the third-order derivatives will stabilize the oscillations right above the sharpness limit**.
- Wu et al. (2024) completely **characterized GD's dynamics** on the binary logistic regression with linearly-separable data (+ regularization (Wu et al., 2025)).

Our Research Question: Can we characterize the convergence behavior of GD with a large stepsize from the binary classification to the multi-class classification setting, and does it exhibit a similar phased convergence pattern as in the binary case?

Contents

Multinomial Logistic Regression (MLR)

Problem Definition

Multinomial Logistic Regression (Böhning, 1992)

Given a dataset with n samples $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$ with $\mathbf{x}^{(i)} \in \mathbb{R}^d$ and $y^{(i)} \in [K]$, we consider the task of predicting the class label $y^{(i)}$ with $\mathbf{x}$. Let $\mathbf{W} = [\mathbf{w}_1^\top; \mathbf{w}_2^\top; \ldots; \mathbf{w}_K^\top] \in \mathbb{R}^{K \times d}$ be the model's weight matrix, where each row $\mathbf{w}_k^\top \in \mathbb{R}^{1 \times d}$ represents the weight vector for class k. Then, for an input $\mathbf{x} \in \mathbb{R}^d$, the model first computes the logit vector, and then computes the probability that it belongs to class k using the softmax function

$$\mathbf{o} = \mathbf{W}\mathbf{x}, \widehat{\mathbf{y}} = \mathsf{softmax}(\mathbf{o}),$$

and

$$\widehat{y}_k = \frac{\exp(\mathbf{o}_k)}{\sum_{j=1}^{K} \exp(\mathbf{o}_j)} = \frac{\exp(\mathbf{w}_k^\top \mathbf{x})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j^\top \mathbf{x})}, \text{ where } k = 1, 2, \ldots, K.$$

Problem Definition (cont.)

Multinomial Logistic Regression (Böhning, 1992)

To find the optimal weight matrix $\mathbf{W}$ based on the given dataset, we find a minimizer of the standard cross-entropy loss function $\mathcal{L} : \mathbb{R}^{K \times d} \mapsto \mathbb{R}$, which is defined as

$$\mathcal{L}(\mathbf{W}) := \frac{1}{n} \sum_{i=1}^{n} \ell_i(\mathbf{W}) = \frac{1}{n} \sum_{i=1}^{n} \ell(\boldsymbol{\delta}^{(i)}, \widehat{\mathbf{y}}^{(i)}),$$

where

$$\ell(\boldsymbol{\delta}^{(\boldsymbol{i})}, \widehat{\mathbf{y}}^{(i)}) = -\sum_{k=1}^{K} \delta_k^{(i)} \log \widehat{y}_k^{(i)},$$

and $\boldsymbol{\delta}^{(i)}$ is the one-hot vector of the label $y^{(i)}$ associated with the i-th feature vector $\mathbf{x}^{(i)}$.

Problem Definition (cont.)

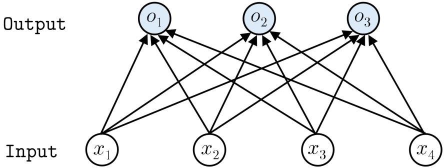


Figure: Multinomial regression can also be seen as a single-layer neural network. Structurally, this building block is frequently found in the last layer of a deep neural network architecture for classification tasks.

Problem Definition (cont.)

Formally, define the optimization problem

$$\min_{\mathbf{W}\in\mathbb{R}^{K\times d}} \mathcal{L}(\mathbf{W}) = -\frac{1}{n}\sum_{i=1}^{n} \log \widehat{y}^{(i)}_{y^{(i)}}. \qquad (\mathcal{P})$$

$\mathcal{L}(\mathbf{W})$ is a convex function and $(\mathcal{P})$ is a convex optimization problem. Moreover, $(\mathcal{P})$ does not have a finite minimizer when the data is linearly separable. Indeed, suppose that $\mathbf{W}_\star$ separates the dataset, then $\log \widehat{y}^{(i)}_{y^{(i)}}|_{\mathbf{W}=c\cdot\mathbf{W}_\star} \to 0$ for all $i \in [n]$ and $\mathcal{L}(c\cdot\mathbf{W}_\star) \to 0$ as $c \to \infty$.
The gradient of the total loss with respect to the weight vector $\mathbf{w}_j$, is

$$\nabla_{\mathbf{w}_j}\mathcal{L}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{n}(\widehat{y}^{(i)}_j - \delta^{(i)}_j)\mathbf{x}^{(i)}.$$

Moreover, the Hessian of the loss function is

$$\mathbf{H}(\mathbf{W}) = \frac{1}{n}\sum_{i=1}^{n}(\mathrm{diag}(\widehat{\mathbf{y}}^{(i)}) - \widehat{\mathbf{y}}^{(i)}\widehat{\mathbf{y}}^{(i)\top}) \otimes \mathbf{x}^{(i)}\mathbf{x}^{(i)\top}.$$

Boundedness and Separability Assumptions

Assumptions on the Dataset

A1 (bounded feature norms). $\forall i \in [n]$, $\|\mathbf{x}^{(i)}\| \leq 1$.
A2 (separability). There exists margin $\gamma > 0$ and

$$\mathbf{W}_* = [\mathbf{w}_{*,1}, \mathbf{w}_{*,2}, \ldots, \mathbf{w}_{*,K}]$$

with $\|\mathbf{W}_*\|_F \leq 1$ such that

$$(\mathbf{w}_{*,y^{(i)}} - \mathbf{w}_{*,k})^\top \mathbf{x}^{(i)} \geq \gamma, \forall i = 1, \ldots, n \text{ and } k \neq y^{(i)}.$$

Our Contribution

Research Question: Can we characterize the convergence behavior of GD with a large stepsize from the binary classification to the multi-class classification setting, and does it exhibit a similar phased convergence pattern as in the binary case?
Our Answer: **Yes**, and we are able to discover several intriguing facts.

Problem	Connection to Perceptron Alg.	EoS Avg. Loss	Transition Time	Stable Phase	Max-Margin
Binary Logistic Regression	✓, Batch variant (Tyurin, 2025)	$\mathcal{O}\left(\frac{1+\eta^2+\ln^2(\eta t)}{\eta t}\right)$ (Wu et al., 2024)	$\mathcal{O}\left(\max\left\{\eta, n, \frac{\eta+n}{\eta}\ln\frac{\eta+n}{\eta}\right\}\right)$ (Wu et al., 2024)	$\widetilde{\mathcal{O}}(1/(\eta t))$ (Wu et al., 2024)	✓(large stepsize: ?) (Soudry et al., 2018)
Multinomial Logistic Regression	✓, Batch variant (Our work)	$\mathcal{O}\left(\frac{1+\eta^2+\ln^2(\eta t K)}{\eta t}\right)$ (Our work)	$\mathcal{O}\left(\max\left\{\eta, n, \frac{\eta+n}{\eta}\ln\frac{(\eta+n)K}{\eta}\right\}\right)$ (Our work)	$\mathcal{O}(1/(\eta t))$ (Our work)	✓(any stepsize) (Soudry et al. (2018)+Our work)

Table: Theoretical contributions: Extending large-stepsize gradient descent analysis from binary to multinomial logistic regression. Our work establishes connection to the perceptron algorithm, convergence bounds for the EoS phase, phase transition time, and stable phase, revealing a mild logarithmic dependence on the number of classes K, and generalization to the max-margin direction.

Our Challenge

From binary classification (only one weight vector) to K-class classification...

- **Complex Gradient Coupling.** Each gradient update pushes the correct class but also pulls all competing classes back.
- **Geometric Degradation of Margins.** As K increases, γ shrinks as $\mathcal{O}(1/\sqrt{K})$, under a fixed norm budget $\|\mathbf{W}_\star\|_F \leq 1$.
- **Normalization and Scaling Sensitivity.** Proper feature normalization dramatically improves convergence in binary classification. Also need a careful choice of norm bounding the optimal weight vector.

Intuition: Reduction to the Batch Perceptron Algorithm

The Multi-Class Batch Perceptron Algorithm

Require: Dataset $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$ where $\mathbf{x}^{(i)} \in \mathbb{R}^d$, $y^{(i)} \in \{1, \ldots, K\}$, Initialization $\mathbf{W}^{(0)}$

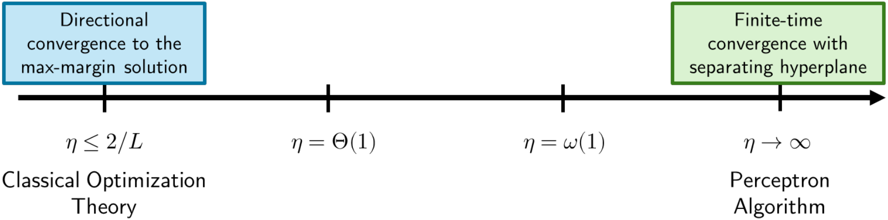
Ensure: Weight matrix $\widehat{\mathbf{W}} \in \mathbb{R}^{K \times d}$ that separates the data

for $t = 0$ to T **do**
    $\mathcal{S}^{(t)} \leftarrow \emptyset$ ▷ Set of misclassified examples
    for $i = 1$ to n **do**
        $\mathcal{M}_i^{(t)} \leftarrow \{k : \widehat{\mathbf{w}}_k^{(t)\top} \mathbf{x}^{(i)} = \max_{k'} \widehat{\mathbf{w}}_{k'}^{(t)\top} \mathbf{x}^{(i)}\}$ ▷ Set of max-scoring classes
        $\widehat{\mathbf{p}}^{(i)} \leftarrow \mathbf{0}$ ▷ Initialize prediction vector
        for $k \in \mathcal{M}_i^{(t)}$ **do**
            $\widehat{p}_k^{(i)} \leftarrow 1/|\mathcal{M}_i^{(t)}|$ ▷ Uniform distribution over tied classes
        end for
        if $\widehat{\mathbf{p}}^{(i)} \neq \boldsymbol{\delta}^{(i)}$ **then**
            $\mathcal{S}^{(t)} \leftarrow \mathcal{S}^{(t)} \cup \{i\}$ ▷ Add to update set if prediction $\neq$ the true label
        end if
    end for
    if $\mathcal{S}^{(t)} = \emptyset$ **then**
        break ▷ Perfect classification achieved
    end if
    $\boldsymbol{\Delta}^{(t)} \leftarrow \frac{1}{n} \sum_{i \in \mathcal{S}^{(t)}} (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}) \mathbf{x}_i^\top$
    $\widehat{\mathbf{W}}^{(t+1)} \leftarrow \widehat{\mathbf{W}}^{(t)} + \boldsymbol{\Delta}^{(t)}$ ▷ Update all classes simultaneously
end for
return $\widehat{\mathbf{W}}^{(t)}$

Our Results

Theorem (Batch Perceptron Reduction)

Under Assumptions A1, A2, and for any dataset, GD on multinomial logistic regression reduces to the Multiclass Batch Perceptron Algorithm, i.e. $\mathbf{W}^{(t)}/\eta \to \widehat{\mathbf{W}}^{(t)}$ for all $t \geq 0$, given that $\mathbf{W}^{(0)} = \mathbf{O}_{K\times d}$ and $\eta \to \infty$.



An Intuitive (and Correct) Prediction: GD should maintain good convergence properties for moderately large η.

Our Results

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Proof Sketch. Dividing both sides from the GD update by η yields

$$\frac{\mathbf{W}^{(t+1)}}{\eta} = \frac{\mathbf{W}^{(t)}}{\eta} + \frac{1}{n} \sum_{i=1}^{n} (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{y}}^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}}) \mathbf{x}^{(i)\top}.$$

Let $\mathcal{M}_i^{(t)} \leftarrow \{k : \widehat{\mathbf{w}}_k^{(t)\top} \mathbf{x}^{(i)} = \max_{k'} \widehat{\mathbf{w}}_{k'}^{(t)\top} \mathbf{x}^{(i)}\}$. We analyze the limit as $\eta \to \infty$, $\max_j \exp(\mathbf{w}_j^\top \mathbf{x}^{(i)})$ dominates, $\widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}} \to 1/|\mathcal{M}_i^{(t)}|$ if $j \in \mathcal{M}_i^{(t)}$; Otherwise, $\widehat{y}_j^{(i)}|_{\mathbf{W}=\mathbf{W}^{(t)}} \to 0$. Hence, the GD update (divided by η) resembles with the perceptron update term $(1/n) \sum_{i=1}^{n} (\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}}) \mathbf{x}^{(i)\top}$. $\square$

Our Results

Theorem (Convergence of Batch Perceptron)

Under Assumption A1, and that $\|\mathbf{x}^{(i)}\| \leq R$ for every $i \in [n]$, the batch perceptron algorithm terminates in at most $8nR^2/\gamma^2$ steps.

Proof Sketch. Consider the potential function

$$\|\widehat{\mathbf{W}}^{(t+1)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 = \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 + 2\langle\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star, \mathbf{\Delta}^{(t)}\rangle_F + \|\mathbf{\Delta}^{(t)}\|_F^2,$$

where

$$\mathbf{\Delta}^{(t)} = \frac{1}{n}\sum_{i\in\mathcal{S}^{(t)}}(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}})\mathbf{x}_i^\top.$$

$$\begin{aligned}\langle\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star, \mathbf{\Delta}^{(t)}\rangle &= \frac{1}{n}\sum_{i\in\mathcal{S}^{(t)}}\left[(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)})^\top\widehat{\mathbf{W}}^{(t)}\mathbf{x}^{(i)} - \alpha(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)})^\top\widehat{\mathbf{W}}_\star\mathbf{x}^{(i)}\right] \\ &\geq -\frac{\alpha\gamma|\mathcal{S}^{(t)}|}{2n}.\end{aligned}$$

Our Results

Theorem (Convergence of Batch Perceptron)

Under Assumption A1, and that $\|\mathbf{x}^{(i)}\| \le R$ for every $i \in [n]$, the batch perceptron algorithm terminates in at most $8nR^2/\gamma^2$ steps.

Proof Sketch.

$$\left\|\boldsymbol{\Delta}^{(t)}\right\|_F^2 = \left\|\frac{1}{n}\sum_{i\in\mathcal{S}^{(t)}}\left(\boldsymbol{\delta}^{(i)} - \widehat{\mathbf{p}}^{(i)}|_{\widehat{\mathbf{W}}=\widehat{\mathbf{W}}^{(t)}}\right)\mathbf{x}^{(i)\top}\right\|_F^2$$

$$\le \frac{1}{n^2}\left(\sum_{i\in\mathcal{S}^{(t)}}\sqrt{2}\cdot R\right)^2 = \frac{2R^2|\mathcal{S}^{(t)}|^2}{n^2}.$$

Let $\alpha = 4R^2/\gamma$, then

$$\|\widehat{\mathbf{W}}^{(t+1)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 \le \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 - \frac{2R^2|\mathcal{S}^{(t)}|}{n} \le \|\widehat{\mathbf{W}}^{(t)} - \alpha\widehat{\mathbf{W}}_\star\|_F^2 - \frac{2R^2}{n},$$

since $|\mathcal{S}^{(t)}| \ge 1$.

Our Results

Theorem (Convergence of Batch Perceptron)

Under Assumption A1, and that $\|\mathbf{x}^{(i)}\| \leq R$ for every $i \in [n]$, the batch perceptron algorithm terminates in at most $8nR^2/\gamma^2$ steps.

Since $\mathbf{W}^{(0)} = \mathbf{W}_{K\times d}$, $\|\mathbf{W}^{(0)} - \alpha\mathbf{W}_\star\|_F^2 = \alpha^2$, T iterations are sufficient as we need to guarantee that

$$0 \leq \|\mathbf{W}^{(T)} - \alpha\mathbf{W}_\star\|_F^2 \leq \alpha^2 - T \cdot \frac{2R^2}{n}$$

and hence

$$T \leq \frac{\alpha^2 n}{2R^2} = \frac{(4R^2/\gamma)^2 n}{2R^2} = \frac{8nR^2}{\gamma^2}.$$

□

Our Results

Theorem (Convergence of Batch Perceptron)

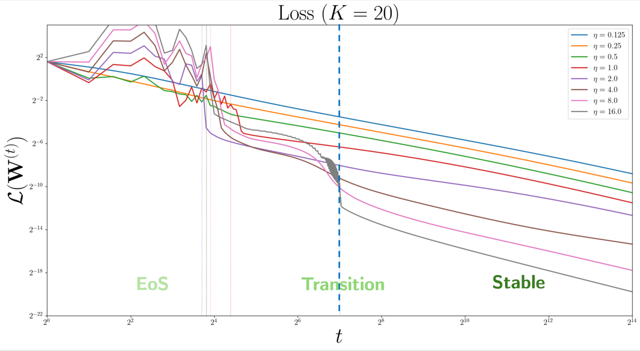
Under Assumption A1, and that $\|\mathbf{x}^{(i)}\| \leq R$ for every $i \in [n]$, the batch perceptron algorithm terminates in at most $8nR^2/\gamma^2$ steps.

An Improvement, and Trade-Off:

- The modification over tied class **eliminates the need for non-degeneracy assumption** (*cf.* Assumption 3.1 in Tyurin (2025)).
- Assuming that there are no tied classes throughout, then steps needed for termination: $\boldsymbol{8nR^2/\gamma^2 \to 2nR^2/\gamma^2}$.

Optimization Dynamics of MLR

Optimization Dynamics Characterization



Training process in two phases:

- **EoS phase**: non-monotonic loss with declining averaged loss.
- **Stable phase**: loss decreases monotonically, subsequently converge at a rate of $\mathcal{O}(1/t)$.

Our Results

Theorem (Optimization Dynamics of GD with MLR on Separable Dataset).

EoS phase. Under Assumptions A1-A2, for every $\eta > 0$, $\mathbf{W}_0 = \mathbf{0}$, we have

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathcal{L}(\mathbf{W}^{(t)}) \leq \frac{\ln^2(\gamma^2\eta T(K-1)) + \eta^2 + 1}{\eta\gamma^2 T},$$

Stable phase. Suppose that there exists s such that

$$\mathcal{L}(\mathbf{W}^{(s)}) \leq \frac{1}{\eta},$$

then $(\mathcal{L}(\mathbf{W}^{(t)}))_{t\geq s}$ decreases monotonically. and

$$\|\mathbf{W}^{(T)}\|_F \leq \frac{\sqrt{2} + 2\ln(\gamma^2\eta T(K-1)) + 2\eta}{\gamma}.$$

Our Results

Theorem (Optimization Dynamics of GD with MLR on Separable Dataset).

Transition time. Let

$$\tau := \frac{96}{\gamma^2} \max\left\{n, \eta, e, \frac{\eta+n}{\eta} \ln \frac{(\eta+n)(K-1)}{\eta}\right\},$$

then there exists $0 \le s' \le 2\tau$ such that

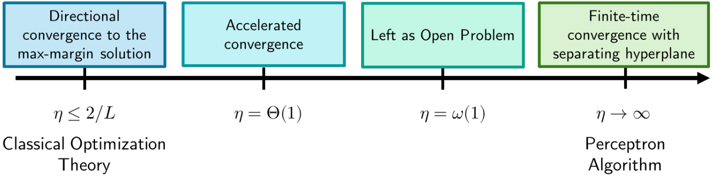
$$\mathcal{L}(\mathbf{W}^{(s')}) \le \min\left\{\frac{1}{2\eta}, \frac{1}{2n}\right\}, G(\mathbf{W}^{(s')}) \le \frac{1}{2n}.$$

Stable descent. For every $t \ge s'$ we have

$$\mathcal{L}(\mathbf{W}^{(t)}) \le \frac{8}{\eta\gamma^2(t-s')}.$$

Remarks

1. **The transition time bound $\tau = \mathcal{O}(\max\{n, \eta\}/\gamma^2)$ corroborates with the Perceptron convergence bound ($\mathcal{O}(n/\gamma^2)$ iterations required).** We can choose a moderate stepsize $\eta := \mathcal{O}(n)$ to match with the perceptron algorithm.



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1. **The transition time bound $\tau = \mathcal{O}(\max\{n, \eta\}/\gamma^2)$ corroborates with the Perceptron convergence bound ($\mathcal{O}(n/\gamma^2)$ iterations required).** We can choose a moderate stepsize $\eta := \mathcal{O}(n)$ to match with the perceptron algorithm.
2. GD with a larger stepsize takes more iterations to exit the initial EoS phase, but it also converges quicker in the stable phase.

Corollary (Acceleration of GD by Large Stepsize).

Under Assumptions A1-A2, consider running GD with a given budget of T steps, where $T \geq 192 \cdot \max\{n, e, \ln(2K-2)\}/\gamma^2$. Then selecting $\eta := \gamma^2 T/192$ guarantees that the transition time τ is sufficient for GD to enter the stable phase, satisfies $\tau \leq T/2$, and

$$\mathcal{L}(\mathbf{W}^{(T)}) \leq \frac{3072}{\gamma^4 T^2}.$$

Remarks

3. **Proof Technique**: **Split optimization bound (Wu et al., 2024):**

$$\begin{aligned}\|\mathbf{W}^{(t+1)} - \mathbf{U}\|_F^2 &= \|\mathbf{W}^{(t)} - \mathbf{U}\|_F^2 - 2\eta\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{W}^{(t)} - \mathbf{U}\rangle_F \\ &\quad + \eta^2\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2 \\ &\leq \|\mathbf{W}^{(t)} - \mathbf{U}\|_F^2 + 2\eta(\mathcal{L}(\mathbf{U}_1) - \mathcal{L}(\mathbf{W}^{(t)})) \\ &\quad + \eta(2\langle\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)}), \mathbf{U}_2\rangle_F + \eta\|\nabla_{\mathbf{W}^{(t)}}\mathcal{L}(\mathbf{W}^{(t)})\|_F^2).\end{aligned}$$

4. **Careful choice of $\mathbf{U}_1$** is the main reason for the **weak logarithmic dependence** on the number of classes K.
5. A surrogate potential function:

$$G(\mathbf{W}^{(t)}) := \frac{1}{n}\sum_{i=1}^{n}(1 - \widehat{y}^{(i)}_{y^{(i)}})$$

6. The Hessian self-boundedness inequality $\|\nabla^2\mathcal{L}(\mathbf{W})\|_{\mathsf{op}} \leq 2\mathcal{L}(\mathbf{W})$ requires the observation of the block Hessian structure to apply Gershgorin Circle Theorem, thence removed the polylog factor in the stable descent bound as **an improvement (also answers Wu et al. (2024)'s question on asymptotics)**.

Implicit Bias of GD on MLR

Our results

Theorem (Margin Maximization of GD on MLR).

For almost all linearly separable datasets, the iterates generated by GD with **any constant stepsize** $\eta > 0$, $(\mathbf{W}^{(t)})_{t \geq 0}$, converge to the max-margin direction, i.e.

$$\lim_{t \to \infty} \frac{\mathbf{w}_k^{(t)}}{\|\mathbf{W}^{(t)}\|_F} = \frac{\mathbf{w}_{\star,k}}{\|\mathbf{W}_\star\|_F},$$

for every $k \in [K]$.

The implicit bias toward max-margin solutions is not limited to the conservative optimization regime, but **persists even with aggressive stepsizes** that exhibit oscillatory behavior in the loss.

Our results

Theorem (Margin Maximization of GD on MLR).

For almost all linearly separable datasets, the iterates generated by GD with **any constant stepsize** $\eta > 0$, $(\mathbf{W}^{(t)})_{t\geq 0}$, converge to the max-margin direction, i.e.

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for every $k \in [K]$.

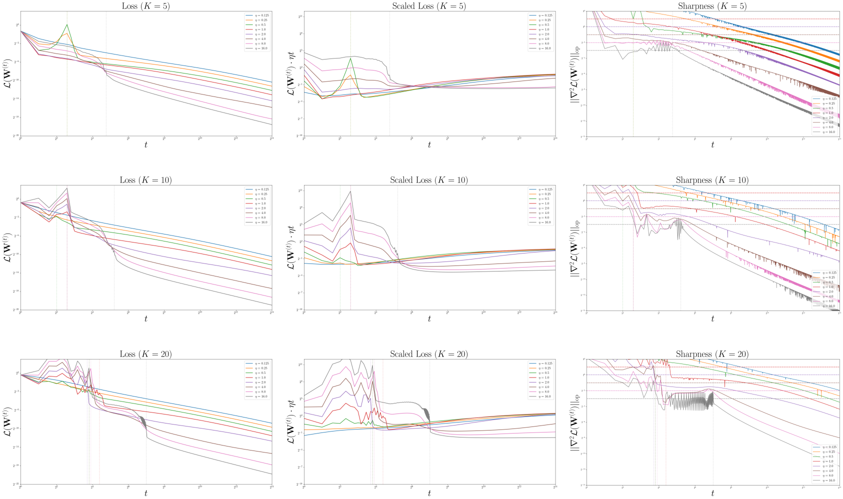
Proof Idea. We patch several prerequisite lemmata in Soudry et al. (2018).

1. Loss convergence implies margin divergence;
2. Margin divergence implies weight divergence;
3. Square-summable gradient norms due to the optimization dynamics;
4. Apply the (patched) Theorem 7 of Soudry et al. (2018).

Experiments

Synthetic Dataset with `make_classification`

We use `make_classification` and `scikit-learn` with $n = 1,000, d = 12, T = 16,384, \texttt{class\_sep} = 4.0$.



Takeaways

- The **sharpness oscillates slightly above** the horizontal lines (corresponding to $2/\eta$) in the EoS phase (*cf.* Cohen et al. (2021)).
- $\mathcal{L}(\mathbf{W}^{(t)}) \cdot \eta t = \mathcal{O}(1)$ and indeed **stabilizes to a constant empirically**.
- Last iteration for loss to oscillate is a good estimate for the transient time.

Ablation Study I

Question: Suppose that the inter-class distance between clusters is fixed, how does γ vary with K? Can we empirically verify how the number of classes K affect the GD dynamics?

Claim (Not Rigorous). The K-dependence comes from:

- **Dimensional Competition**: With K classes, there are $K-1$ "pull" directions competing against one "push" direction. As there are more classes, there are also more ways for the gradient updates to interfere with each other geometrically. $\rightarrow$ Empirically explained by the **logarithmic dependence**.
- **The Margin γ**: For a fixed inter-distance (between clusters) dataset, as K increases, the max-margin solution $\mathbf{W}_\star$ has to satisfy more constraints. To keep its Frobenius norm bounded by 1, the "importance" assigned to any single class direction must generally decrease. This leads to a smaller achievable margin γ for the same set of data points as K increases. $\rightarrow$ **Let's try to control γ explicitly...**

Our Dataset Construction Method

1. We set $d = 12$, number of samples $n = 5,000$, and margin target $\gamma = 0.0004$.
2. Randomly sample K ground-truth vectors $\mathbf{w}_j \in \mathbb{R}^d$, and scale to ensure that $\|\mathbf{W}_\star\|_F = 1$.
3. Sample n data points with random norms in the range $[0.01, 20]$ to explicitly construct a dataset with the target margin γ with rejection sampling. Accept the sample if $\left(\mathbf{w}_{\star,y} - \mathbf{w}_{\star,k}\right)^\top \mathbf{x} \geq \gamma$ for all $k \neq y$ ($\gamma \approx 0.0004$).

Experimental Results

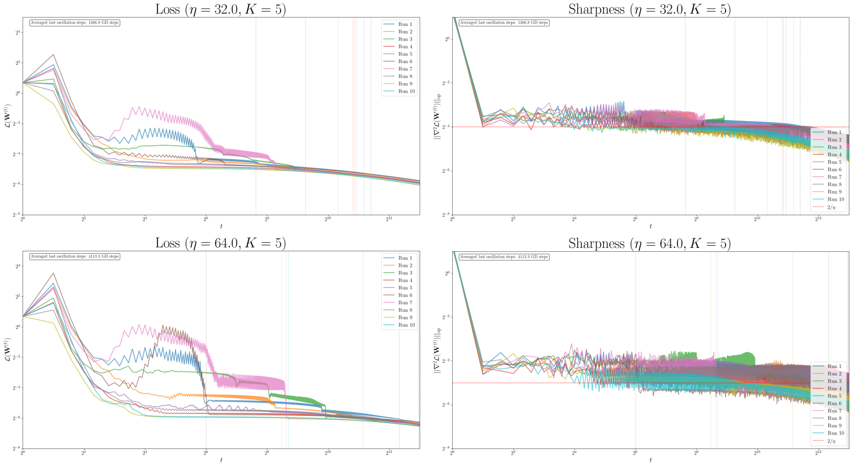


Figure: Results of the first ablation experiment, where $K = 5$.

Experimental Results

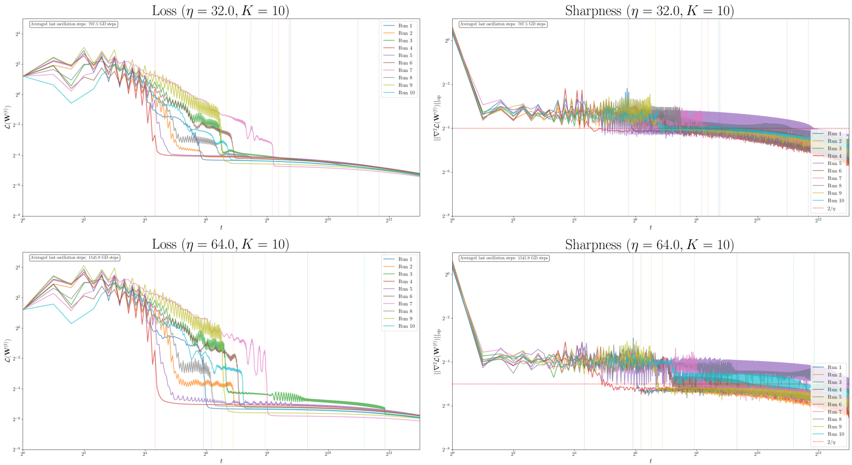


Figure: Results of the first ablation experiment, where $K = 10$.

Experimental Results

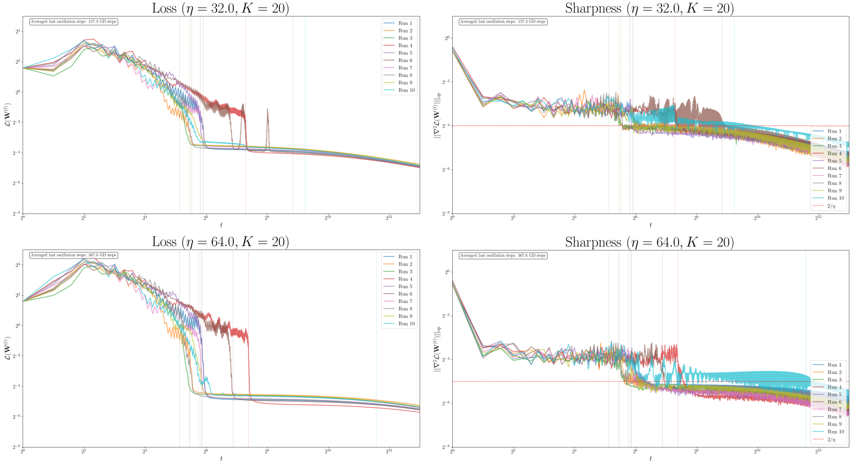


Figure: Results of the first ablation experiment, where $K = 20$.

Ablation Study II

Fixed number of data points for each class.

- Let the margin γ scale accordingly with K to match with the transition time bound $\mathcal{O}(n/\gamma^2)$.
- Keeping this factor constant, we have $\gamma_K = \gamma/\sqrt{K}$

Experimental Results

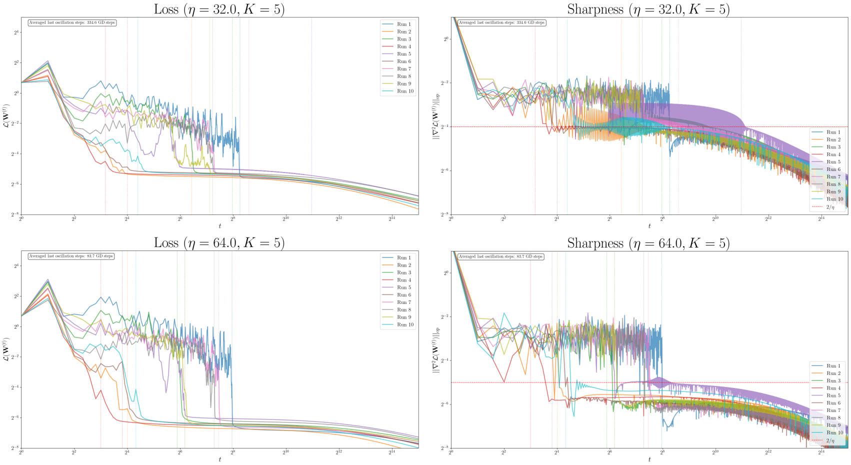


Figure: Results of the second ablation experiment, where $K = 5$.

Experimental Results

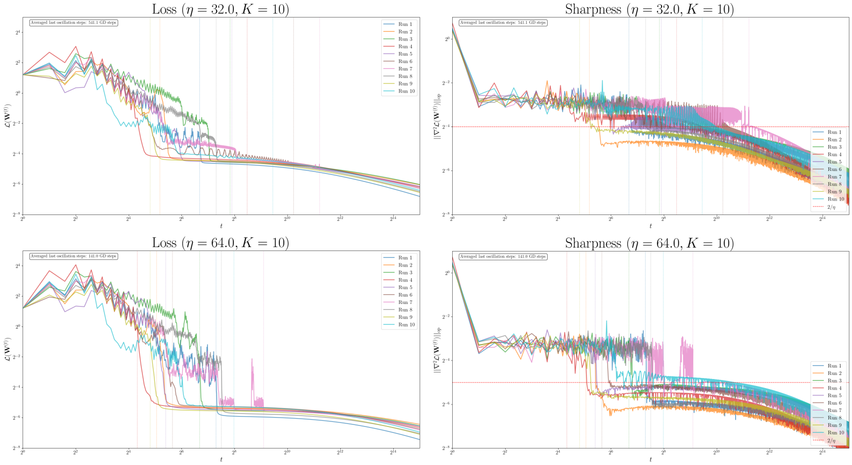


Figure: Results of the second ablation experiment, where $K = 10$.

Experimental Results

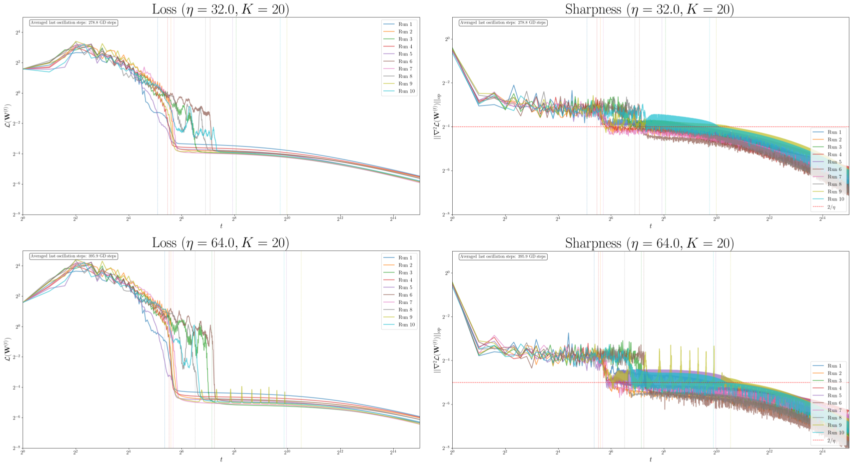
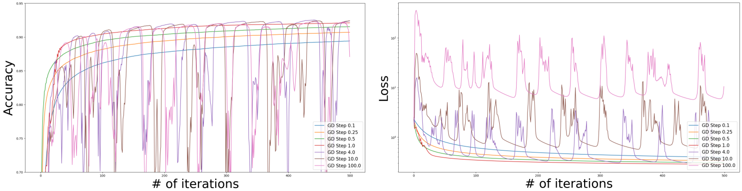


Figure: Results of the second ablation experiment, where $K = 20$.

Future Works

Multi-class Softmax Regression: Non-separable Dataset

Softmax classification on MNIST dataset. Strong oscillation persists.



Large stepsize could fail... This is possibly because:

- Samples are non-linearly separable.
- Ahn et al. (2022): GD dynamics cannot converge to stationary point with sharpness greater than $2/\eta$ ($\lambda_{\max}(\nabla^2\mathcal{L}(\mathbf{W})) > 2/\eta$), need to ensure $\eta \leq 4$ (if $\|\mathbf{x}\| \leq 1$).

Some Promising Directions

1. SGD on MLR with separable dataset
2. Beyond the linear classifiers: NTK regime...?
3. Sharpness-aware algorithm design and analysis
4. Adding regularizations...
5. Non-separable dataset (will be carried out in Term 2)

Conclusion

1. GD on MLR with separable dataset - **accelerated convergence with oscillations, but not catastrophic**.
2. **Comparison with batch perceptron algorithm** - additional GD steps earns convergence to *max-margin* solution.
3. Implicit bias of GD is over the **entire stepsize spectrum**.
4. Can be served to **explain some of the unstable convergence effect in modern DL practices**.
5. **Stemmed from the main texts (along with some improvements):**
   Tyurin (2025) → removed data-degeneracy assumptions
   Wu et al. (2024) → multi-class
   Soudry et al. (2018) → small stepsize is unnecessary.

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Thank you!